[*Working title, to be decided*]

Anonymous ACL submission

Abstract

[*Abstract to be written after the results section.*]

1 Introduction

When many users share a resource whose cost rises with use, each user's choice slows down the others. Roads, communication networks, power grids and compute clusters all have this structure, and game theory models it as a congestion game (Rosenthal, 1973). Language-model agents that act for people will often share such resources: they queue jobs on shared accelerators, call rate-limited services, book delivery slots and charging stations, and choose routes. We use the congestion game in this general sense, as a model of any resource that agents share, and not as a model of traffic.

Congestion games separate two standards of good play. If each user minimizes its own cost, play settles where no user gains by switching alone, the user equilibrium; the system optimum instead minimizes the total cost (Wardrop, 1952). The two generally differ, and the ratio between them is the price of anarchy (Roughgarden and Tardos, 2002). Consider four users and two routes, one whose travel time grows with the number of users on it and one with a fixed travel time (Pigou, 1920). The congestible route is never slower than the alternative, so self-interested users all take it and each pays the same as on the alternative; the total cost is 4, whereas sending two users each way costs 3. This system optimum is unequal: the two users on the slower route pay twice as much as the other two. Repetition removes the conflict. Over eight trips, keeping the same two users on the slower route gives cumulative costs of 4, 4, 8 and 8, while taking turns gives 6 to everyone, and both are optimal on every trip. In repeated play, efficiency and an equal division of its cost can be achieved together, so who takes the slow road is a choice among equally efficient outcomes and not a trade-off against efficiency.

Making that choice requires users who face the same situation to act differently. Congestion games reward being in the minority, and identical reasoning defeats itself: if everyone concludes that the less crowded route is better, everyone takes it. In models of the El Farol bar problem and the minority game, populations that share one strategy crowd and oscillate, and efficiency depends on a diversity of strategies (Arthur, 1994; Challet and Zhang, 1997). In laboratory route-choice games, human groups keep fluctuating around equilibrium (Selten et al., 2007) and discover alternation only after hundreds of rounds (Helbing et al., 2005). Game theory adds that players with identical strategies cannot coordinate in a symmetric game unless something tells them apart (Crawford and Haller, 1990). A group of language-model agents built from the same model and the same prompt is the limiting case of identical reasoners. Work on algorithmic monoculture, the reliance of many decision makers on one model, expects such groups to fail together (Kleinberg and Raghavan, 2021; Bommasani et al., 2022).

Unlike the players in these models, language-model agents can talk. Non-binding messages before play, or cheap talk, can help players select among equilibria (Farrell and Rabin, 1996; Cooper et al., 1992), but theory does not say which equilibrium they select or how. Studies of LLM agents in social and coordination games measure whether cooperation or coordination succeeds (Piatti et al., 2024; Agashe et al., 2025) and find that communication can raise it (Piatti et al., 2024; Madmoun and Lahlou, 2026); independently queried LLMs agree on a common answer when they should match but fail to differ when differing pays (Ballesteros et al., 2026). These studies measure whether coordination succeeds. They do not show how a group of identical agents talking in natural lan-

guage decides who will bear a cost: whether the agents negotiate, how they reach agreement, and whether the agreement holds.

This paper studies that process. We place groups of four LLM agents in the repeated two-route game above, where the efficient and equal outcome exists, and let them exchange declared routes and free-text notes for three rounds before each trip, either shared with the group or kept private. We ask four questions.

- **RQ1.** Without coordination, do LLM groups stay at the user equilibrium? Does an objective of minimizing total cost move them to the system optimum, and does it also settle who bears the cost?
- **RQ2.** Does asking for equal totals spread the cost without a loss of efficiency?
- **RQ3.** What do messages add, and how do the agents reach agreement through them?
- **RQ4.** Where does this coordination break down, and does it depend on the agents sharing a model?

We answer RQ1 and RQ2 with comparisons fixed before data collection, and RQ3 and RQ4 with the same comparisons and an analysis of the messages themselves, across four open-weight models from two families with native reasoning on and off, and one closed model.

Our contributions are threefold. First, we provide a congestion-game testbed in which efficiency and the division of cost can be measured separately, with 8,578 games and their complete message logs. Second, we show that groups of identical agents agree on who pays without negotiating: before reading one another, several agents write the same complete assignment, and the exchange that follows confirms it rather than bargains over it. Third, we identify when this coordination works and when it fails, which recasts monoculture, usually treated as a source of correlated failure, as a coordination resource whose reach is limited to problems with a single, easily re-derived answer.

2 Related Work

[*Three paragraphs: congestion games, route choice and the minority game; LLM agents in social and coordination games; conventions, symmetry breaking and monoculture.*]

3 Study Design

[*Game and benchmarks (user equilibrium, atomic Nash equilibrium, system optimum, fixed and rotating optima); protocol; definition of a monoculture; models and why open weights; Study 1, Study 2 and the GPT-6 Luna replication; outcome and message measures; what is pre-specified and what is exploratory.*]

4 Results

4.1 RQ1: Total-cost objective

[*Study 1.*]

4.2 RQ2: Equality objective

[*Study 2 and Luna, objective contrasts by message condition.*]

4.3 RQ3: Communication

[*What messages add; how agreement forms; how the exchange closes; how agreement persists; how rotation is agreed.*]

4.4 RQ4: Limits

[*Where coordination breaks down; mixed-model groups and first-round probes (planned).*]

5 Discussion

[*Monoculture as a coordination device; agreement without commitment; implications for agents that share congestible resources.*]

Limitations

[*To be written.*]

Ethics and Reproducibility

[*To be written.*]

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